

Ice Cream Parlour Management System (ICPMS)

Frontend Architecture & Requirements

Version: 1.0

Technology Stack:

- React.js
 - Vite
 - Tailwind CSS
 - React Router DOM
 - Axios
 - React Hook Form
 - Recharts
 - React Hot Toast
 - Lucide React
 - Cloudinary Integration
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1. Frontend Objectives

The frontend should provide a professional POS and business management experience for:

- Admin
- Manager
- Cashier
- Inventory Staff

The UI should feel similar to modern ERP and POS systems.

Requirements:

- Fast
 - Mobile Responsive
 - Clean Design
 - Role Based Navigation
 - Real-Time Feedback
 - Easy Product Search
 - Attractive Dashboard
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2. Application Layout Structure

Application Layout:

Header/Navbar



Sidebar Navigation



Main Content Area



Footer

3. Global Layout Components

Navbar

Features:

- Company Logo
 - User Profile
 - Notifications
 - Role Badge
 - Logout Button
-

Sidebar

Menu Items:

Dashboard

Products

Inventory

Orders

Customers

Employees

Offers

Reports

Settings

Logout

Menu visibility depends on role.

Notification Panel

Shows:

- Low Stock Alerts
 - New Orders
 - Offer Expiry Alerts
-

4. Routing Structure

Public Routes

/login

/forgot-password

Protected Routes

/dashboard

/products

/products/create

/products/edit/:id

/inventory

/inventory/add

/orders

/orders/:id

/customers

/customers/:id

/employees

/employees/:id

/offers

/reports

/settings

/profile

5. Authentication Pages

Login Page

Fields:

Email

Password

Remember Me

Features:

- Validation
- Error Handling
- JWT Login
- Redirect Based On Role

Buttons:

Login

Forgot Password

Forgot Password

Fields:

Email

Features:

- Email Validation
 - Reset Link Request
-

6. Dashboard Page

Purpose:

Provide complete business overview.

Cards:

Today's Revenue

Today's Orders

Customers Today

Inventory Status

Monthly Revenue

Low Stock Products

Charts:

Daily Sales

Monthly Sales

Top Products

Revenue Analytics

Tables:

Recent Orders

Low Stock Products

Recent Customers

7. Product Management Module

Products Listing Page

Table Columns:

Image

Name

Category

Price

Stock

Status

Actions

Features:

Search Product

Filter Products

Pagination

Sorting

Export Data

Actions:

View

Edit

Delete

Add Product Page

Fields:

Product Name

Description

Category

Price

Cost Price

Stock Quantity

Tax Percentage

Product Image

Status

Features:

Cloudinary Upload

Validation

Preview Image

Submit

Edit Product Page

Same fields as Add Product

Pre-filled data

Update functionality

Product Details Page

Displays:

Product Image

Product Information

Sales Statistics

Inventory Information

8. Inventory Module

Inventory Listing

Columns:

Item Name

Current Stock

Minimum Stock

Status

Last Updated

Features:

Search

Filter

Pagination

Stock Alerts

Add Inventory Item

Fields:

Item Name

Quantity

Unit

Minimum Quantity

Supplier

Purchase Cost

Inventory Details

Stock History

Stock Adjustments

Consumption Report

Purchase Report

9. Billing / POS Module

Most important screen.

Layout:

Left Side:

Product Search

Categories

Product Grid

Right Side:

Cart

Billing Summary

Payment Section

Features:

Add Product

Remove Product

Increase Quantity

Decrease Quantity

Apply Discount

Calculate Tax

Generate Invoice

Print Bill

Download Invoice

Payment Methods:

Cash

UPI

Card

10. Orders Module

Orders List

Columns:

Order ID

Customer

Amount

Payment Status

Order Status

Date

Features:

Search

Filter

View Details

Update Status

Order Details

Customer Info

Ordered Items

Invoice

Payment Details

Status Timeline

11. Customer Module

Customer Listing

Columns:

Name

Mobile

Email

Orders

Total Spending

Actions

Features:

Search

Pagination

Filters

Add Customer

Fields:

Name

Phone

Email

Address

Birthday

Customer Profile

Displays:

Purchase History

Total Orders

Total Spending

Loyalty Points

Favorite Products

12. Employee Module

Employee Listing

Columns:

Name

Role

Contact

Salary

Status

Actions

Add Employee

Fields:

Name

Email

Phone

Role

Salary

Address

Joining Date

Employee Profile

Attendance

Performance

Salary Details

Assigned Tasks

13. Offers & Discounts Module

Offers List

Columns:

Offer Name

Discount Type

Discount Value

Start Date

End Date

Status

Create Offer

Fields:

Offer Name

Description

Discount Type

Discount Value

Applicable Products

Start Date

End Date

14. Reports Module

Reports Available:

Daily Sales

Weekly Sales

Monthly Sales

Inventory Reports

Customer Reports

Product Reports

Revenue Reports

Features:

Date Filters

Export PDF

Export Excel

Print Report

Charts

Tables

15. Settings Module

Business Information

Store Name

GST Number

Address

Phone Number

Invoice Settings

Tax Settings

User Management

Password Change

16. Reusable Components

Button

Input

Select

Modal

Table

Pagination

Search Bar

Loader

Toast

Badge

Avatar

Card

Chart

Confirmation Dialog

File Upload

17. State Management

Use Context API

Contexts:

AuthContext

ProductContext

InventoryContext

OrderContext

CustomerContext

NotificationContext

18. API Service Layer

Folder:

src/services

Files:

authService.js

productService.js

inventoryService.js

orderService.js

customerService.js

employeeService.js

offerService.js

reportService.js

19. Frontend Folder Structure

src/

components/

layouts/

pages/

dashboard/

products/

inventory/

orders/

customers/

employees/

offers/

reports/

settings/

auth/

contexts/

services/

hooks/

utils/

routes/

assets/

constants/

styles/

App.jsx

main.jsx

20. Responsive Design Requirements

Desktop

Laptop

Tablet

Mobile

Breakpoints:

640px

768px

1024px

1280px

1536px

Requirements:

Collapsible Sidebar

Responsive Tables

Mobile Cards

Touch Friendly Buttons

Adaptive Charts

21. UI Design Principles

Modern SaaS Dashboard Style

Design Goals:

- Clean
- Professional
- Fast
- Minimal
- Business Focused

Color Palette:

Primary:

2563EB

Secondary:

1E293B

Success:

16A34A

Warning:

F59E0B

Danger:

DC2626

Background:

F8FAFC

Cards:

FFFFFF

Typography:

Inter Font Family

Font Weights:

400

500

600

700

22. Frontend Deliverables

The frontend must include:

- ✓ Authentication
- ✓ Dashboard
- ✓ Products Module
- ✓ Inventory Module
- ✓ Billing System
- ✓ Orders Module
- ✓ Customers Module
- ✓ Employees Module
- ✓ Offers Module
- ✓ Reports Module
- ✓ Settings Module

✓ Responsive Design

✓ Cloudinary Upload Integration

✓ Role-Based Navigation

✓ Production Ready UI

End of Document